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Performance Analysis of OFDM, W-OFDM and F-OFDM Under Rayleigh Fading Channel for 5G Wireless Communication

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Abstract - The OFDM multiple carrier modulations are more secure and suitable for a special fading that is known as a frequency selective fading. To minimize the queue in the 5G waveform, it is likewise much important to resize the bandwidth in the OFDM based technology. Consequently, Filtered Orthogonal Frequency Division Multiplexing also known as a spectrally localized waveform technology which is the primary approach. Windowed-OFDM (W-OFDM), where each symbol is categorized by windowed and every symbol is overlapped in the special time domain that is known as the classical OFDM scheme. The traffic scenario is processed by the individual sub-bands. This article presents the comparison of the performance analysis of OFDM, W-OFDM, and F-OFDM systems using BPSK, QAM, 16-PSK, 8-QAM, QPSK, and 16-QAM modulation techniques under Rayleigh fading channel. The main aim of this article is to focus on examining the effective performance of OFDM, W-OFDM, and F-OFDM in terms of PSD, Bit error rate, and signal to noise ratio. The simulated values are analyzed in terms of PSD and BER.

Keywords— OFDM, F-OFDM, W-OFDM, Relay, MIMO, BPSK, QAM, BER, ISI.

I. INTRODUCTION

The demands and anticipations for the 5th generation wireless networks are increasing day by day after many decades of controversy through perseverance and academics. The short and long-range with high-speed radio access is expected through the millimeter-wave by tens of Gbps the lower frequency bands [1]. In this section, the air interface that is known as an underlying waveform must have to be determined. For researchers and engineers, the next-generation wireless networks present the most challenging matters for wireless communication. The primary aim is to promote the LTE functional activity and the data transfer rate within a short span of time so that the increasing data demand can be eliminated [2]. Recently, to enhance the channel capacity researchers propose 10, 16, or 20 antennas based systems which are promising to perform massive MIMO operation [2], [3].In this work, 4-antenna structures were considered for advanced 4×4 MIMO operation for 5G enabled smartphones. For the proper and suitable development of wireless communications, waveforms are considered consistent and fundamental factors. The cyclically prefixed OFDM (CP-OFDM) is the key factor for the success of 4G LTE (Long-Term Evaluation). It is formulated to combat the effect of multipath response by improving spectral efficiency and significantly mitigating ISI (inter-symbol interference). It is also MIMO friendly.

This article attempts to figure out the best performance for 5G applications among these three filters OFDM, F-OFDM and W-OFDM based one BER, PSD, and signal to noise ratio using BPSK, QAM, 16-PSK, 8-QAM, QPSK, and 16-QAM modulation. The OFDM sub-band has been considered and OOB emission has been omitted [3].

A. OFDM

Orthogonal Frequency-Division Multiplexing (OFDM) is the combination of multiple sub-carriers. OFDM process demands the specific N number of individual sub-carriers to transmit the number of data flows from input source to destination output [4],[5]. Every carrier signals are orthogonal to each other. All of these data streams are converted to the special parallel data stream that is known as an N.

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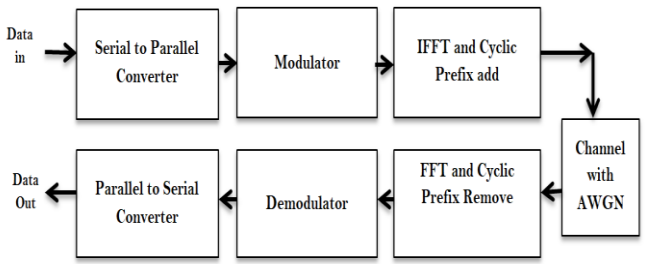


Fig. 1. OFDM Architecture.

To produce a final OFDM signal a special oscillator known as sinusoidal oscillators is totalized together. Fig. 1 above shows the OFDM Architecture [6].

B. *Windowed OFDM* (W-OFDM)

Window filters are also known as the FIR filter, which is used here is to specify the time domain in the windowing strategy. Window filters omits the extra leakage part of the OFDM system and extract the better portion of the FIR signals [7].

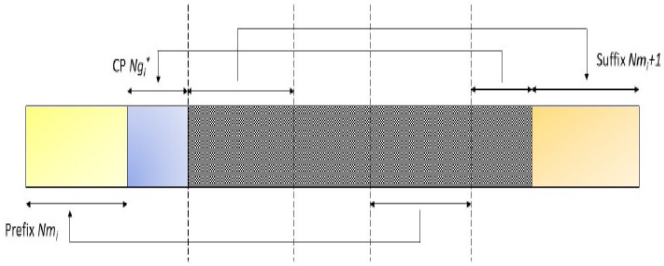


Fig. 2. Prefix, cyclic prefix and suffix development for a W-OFDM symbol.

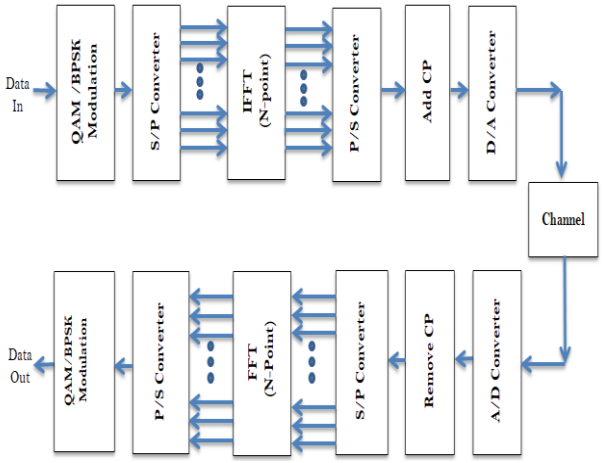


Fig. 3. W-OFDM Architecture.

Initially, the OFDM symbols are extended with the array of cyclic prefixes according to fig.2. W-OFDM an architecture model is denoted by fig.3.

The extended symbol x produces the windowed symbol denoted by x_m via equation (1).

$$x_m = x_i . w_i \quad (1)$$

Where, *the* length $N_i^{(w)}$ is represented by w_i. The window is defined by the equation (2).

$$w_i = \begin{bmatrix} 0^{(N_{mi}-N_{tri/2}).} \\ 1^{(N_i+N_{gi}-N_{tri}+1).} \\ 0^{(N_{mi}-N_{tri/2}).} \end{bmatrix} \quad (2)$$

Where, 0^L represents a special vector known as L vector that is loaded by zeros, likewise 1^L is a similar type of vector filled by ones. The window transition length is represented by the parameter N_{th}. The sub-band of ith produces the transition length that is denoted as N_{ri}.

C. *Filtered OFDM* (F-OFDM)

The processing mechanism and functional property of Filtered-OFDM are similar to the CP-OFDM, only a low-pass filter has been introduced at the end of CP.

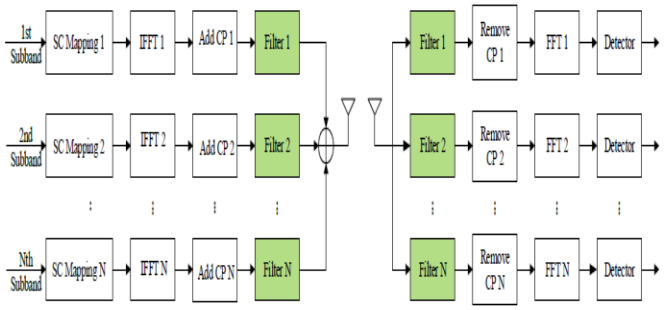


Fig. 4. Downlink transceiver structure of F-OFDM.

The downlink transceiver shape of F-OFDM which is denoted is shown in Fig.1. The OOB emissions are reduced by the possible interference the shape of the transmitter section low-pass filter plays a vital role [8], [9], [10]. A special filter is needed that is not only ideally smooth in pass-band but also zero exterior band.

The Sinc lobe is concentrated to analyze the cyclic prefix during small-time period instead of long-time period. This is the reason for redundancy and the L_i is to be narrow, even L_i can be very broad than L_{gi} [11], [12].

To achieve this specific goal, a windowed-sinc filter is chosen with ideal impulse response

$$p_i(n) = Sinc\left(\Delta f_i (N_{ui} + 2R_i)\, {}^{n}/_{N_i}\right) \quad (3)$$

For $-[L_i/2] \leq n \leq [L_i/2]$, Where L_i represents the filter order and $\Delta f_i R_i$ the transition band in one side [13],[14],[15],[16]. $p_i(n)$ do not represent the final filter, it is only a truncated based sinc. The role of transition bands of the filter is given below in fig.5.

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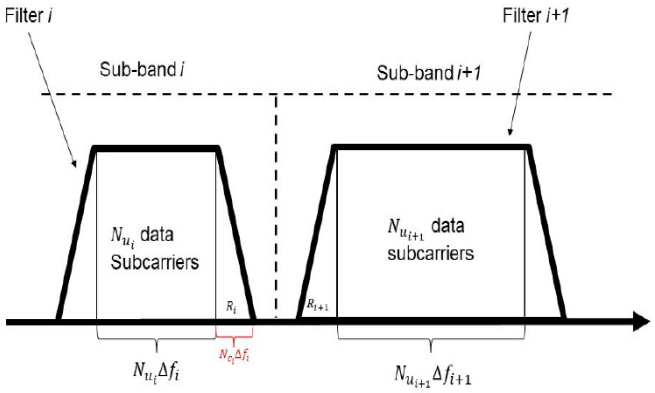


Fig. 5. Role of transition bands of the filter.

The coefficients of the normalized filters are given by the equation (4).

$$f_i(n) = \frac{p_i(n).w_i(n)}{\sum_k p_i(k).w_i(k)} \quad (4)$$

$$w_i(n) = (0.5(1+\cos(\frac{2\pi n}{L_i-1})))^{0.5} \quad (5)$$

Where, the equation is bounded by the n.

II. SYSTEM MODEL

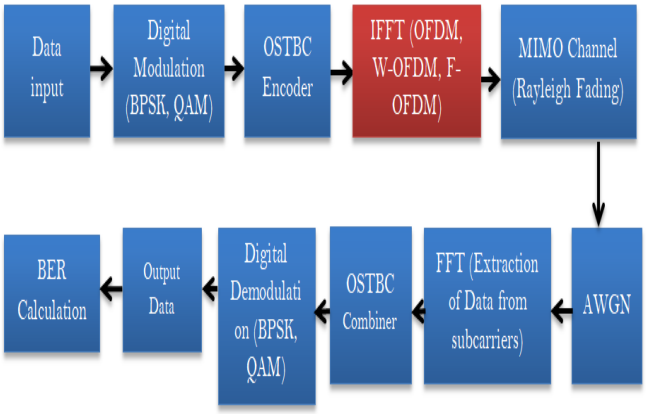


Fig. 6. MIMO incorporated OFDM, WOFDM and F-OFDM wireless systems

The MIMO incorporated OFDM, W-OFDM, and F-OFDM WOFDM systems are modeled for using the OSTBC encoding technique [18],[19],[20],[21]. Initially, the data inputs are modulated by the specific modulation techniques specially BPSK and QAM. The modulated signal will be encoded by the OSTBC encoder [22],[23],[24]. After finishing the encoding process the IFFT will be formed and then passes through the MIMO channel. The Rayleigh fading channels are going through the AWGN and then reconstruct the FFT signals. After reconstructing the signals, a combiner known as an OSTBC combines the FFT signals. Each of the combiner signals transpires through the digital modulation specially BPSK and QAM and then produces the output data.

Finally, after obtaining the output data, calculate the BER for our result and simulation. Total numbers of 192 Samples in each frame have been taken. A total of 100 Bits per symbol are considered for the simulation. W-OFDM and F-OFDM and the symbol period are 10^{-6}s. The system is designed with four (4 x 4) antennas where at the transmission section there will be four antennas and receiving section there will be four antennas.

III. RESULTS AND ANALYSIS

TABLE I
SUMMARY OF SIMULATED MODEL PARAMETERS

Parameter	Considerations for Simulation
Modulation Scheme	BPSK,QPSK,16-PSK,QAM,8QAM, and 16-QAM
Channel	Rayleigh Fading Channel
Multiplexing	OFDM, W-OFDM, F-OFDM
Samples per frame	192
No. of transmitting & receiving antennas	4*4
Signal to Noise Ratio	0 to 25dB

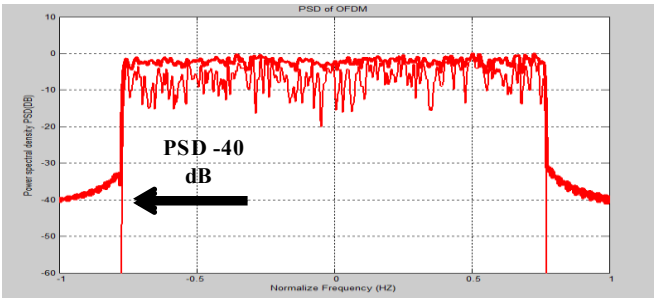


Fig. 7. power spectral density (PSD) of OFDM.

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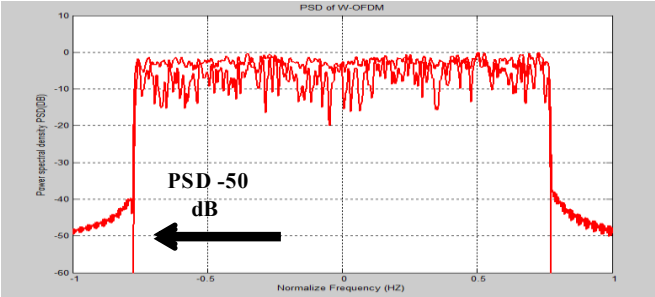


Fig. 8. power spectral density (PSD) of W-OFDM.

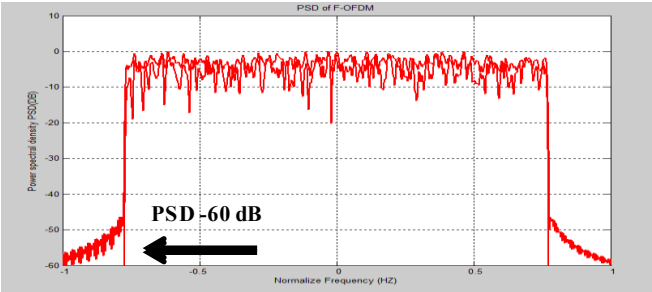


Fig. 9. power spectral density (PSD) of F-OFDM

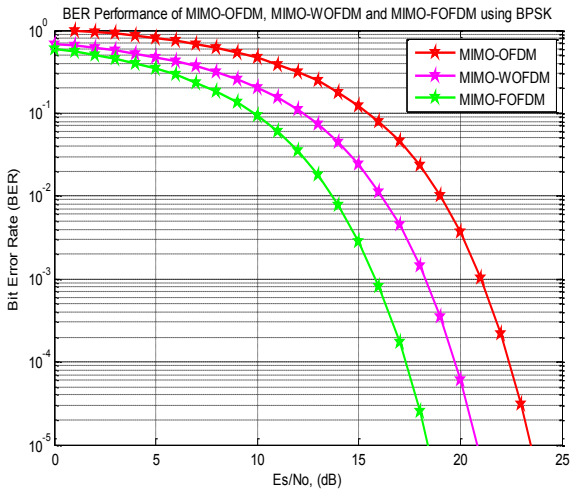


Fig. 10. BER for MIMO-OFDM, MIMO-WOFDM and MIMO-FOFDM using BPSK.

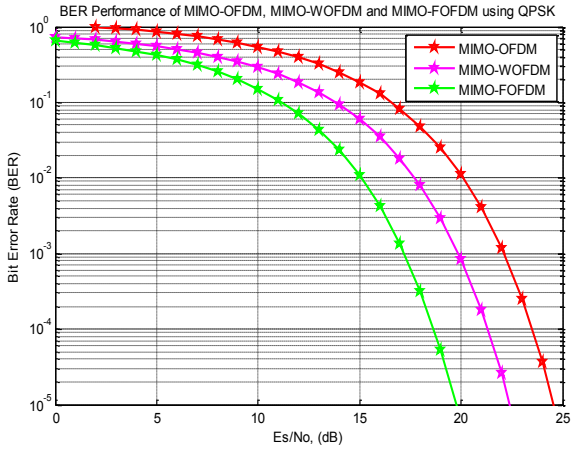


Fig. 11. BER for MIMO-OFDM, MIMO-WOFDM and MIMO-FOFDM using QPSK.

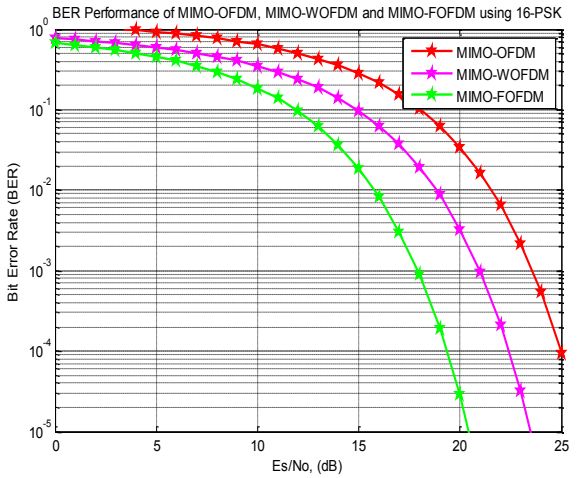


Fig. 12. BER for MIMO-OFDM, MIMO-WOFDM and MIMO-FOFDM using 16-PSK.

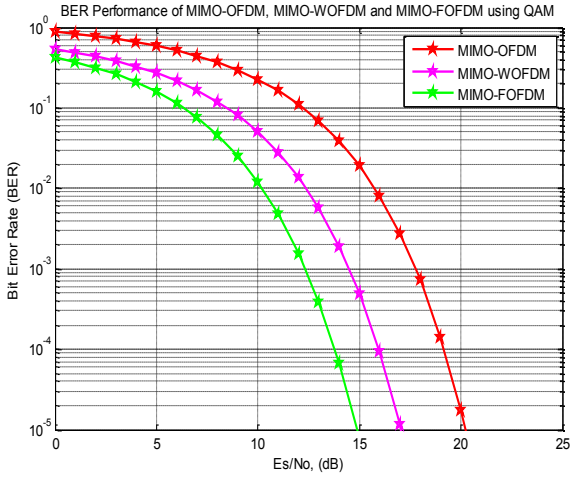


Fig. 13. BER for MIMO-OFDM,MIMO-WOFDM and MIMO-FOFDM using QAM.

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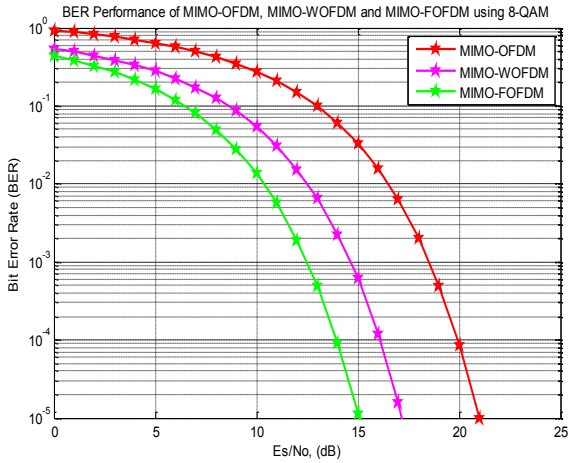


Fig. 14. BER for MIMO-OFDM,MIMO-WOFDM and MIMO-FOFDM using 8-QAM.

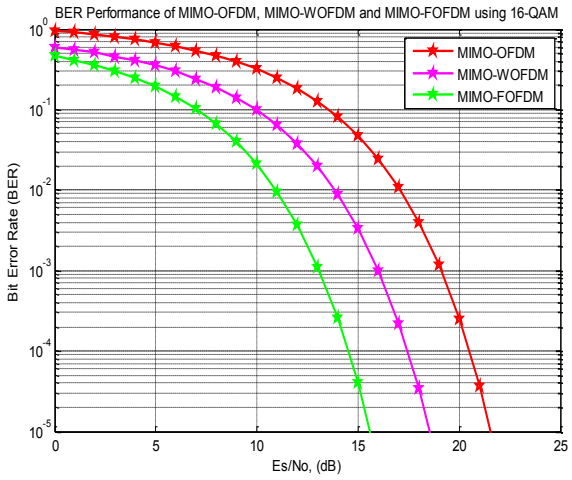


Fig. 15. BER for MIMO-OFDM,MIMO-WOFDM and MIMO-FOFDM using 16-QAM.

Fig.10 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM is 23 dB, 21 dB and 18 dB for Bit Error Rate 10^{-5}. For Fig .11 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM are 24 dB, 23 dB and 19.6 dB for Bit Error Rate 10^{-5}. For Fig.12 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM are 29 dB, 24 dB and 21 dB for BER 10^{-5}. For Fig.13 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM are 21 dB, 17 dB and 14.8 dB for BER 10^{-5}. ForFig.14 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM are 22 dB, 17 dB and 15 dB for Bit Error Rate 10^{-5}. ForFig.15 represents the signal by noise ratio of OFDM, W-OFDM and F-OFDM are 22 dB, 18 dB and 16 dB for BER 10^{-5}. All values are represented in below Table II.

TABLE II
Data table of Modulations, Multiplexing, BER and SNR

Modulation	Multiplexing	BER	SNR
BPSK	OFDM	10^{-5}	23 dB
	W-OFDM	10^{-5}	21 dB
	F-OFDM	10^{-5}	18 dB
QPSK	OFDM	10^{-5}	24 dB
	W-OFDM	10^{-5}	23 dB
	F-OFDM	10^{-5}	19.6 dB
16-PSK	OFDM	10^{-5}	29 dB
	W-OFDM	10^{-5}	24 dB
	F-OFDM	10^{-5}	21 dB
QAM	OFDM	10^{-5}	21 dB
	W-OFDM	10^{-5}	17 dB
	F-OFDM	10^{-5}	14.8 dB
8-QAM	OFDM	10^{-5}	22 dB
	W-OFDM	10^{-5}	17 dB
	F-OFDM	10^{-5}	15 dB
16-QAM	OFDM	10^{-5}	22 dB
	W-OFDM	10^{-5}	18 dB
	F-OFDM	10^{-5}	16 dB

IV. CONCLUSIONS

The performance of OFDM, W-OFDM, and F-OFDM is presented under the Rayleigh fading channel considering Bit Error Rate and Signal to Noise Ratio under different modulation techniques. The PSK higher-order modulation provides high SNR comparing with the admissible latitude of BER in the suitable simulated channel. On the other hand, QAM higher-order modulation provides low SNR comparing with the admissible latitude of BER in the suitable simulated channel. By comparing PSK and QAM modulation techniques, notably, QAM needs low SNR as comparing with PSK for exact BER value. Various parameters were analyzed such as Bit Error Rate, Power Spectral Density, and signal to noise ratio with BPSK, 16-QPSK, PSK, 8-QAM, QAM and 16-QAM modulation. It can be concluded that among three multiplexers F-OFDM provides dominant performance and suitable bandwidth that is efficient for the wireless communication system.

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